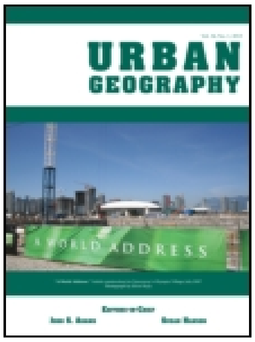

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CLASSIFICATION APPROACHES IN NEIGHBORHOOD RESEARCH: INTRODUCTION AND REVIEW[1]

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COMPLEXITY, THRESHOLDS AND CATEGORIES IN NEIGHBORHOOD RESEARCH

The majority of theoretical questions in neighborhood research center on categorical or interval constructs—phenomena that at least implicitly have distinct interval thresholds and/or boundaries between classes. Such constructs include temporally cross-sectional phenomena such as urban blight, ethnic or other subpopulation enclaves, territories of the underclass, and local niches of particular economic activities as well as dynamic processes such as gentrification, abandonment, and group transitions. The problem this presents to urban scholars is to set boundaries for these intervals and categories when the measures used are predominantly continuous. Formally, the challenge is to properly define and operationalize thresholds: to correctly convert scales of measurement from continuous to ordinal or interval, in ways that conform to theory. Even the simplest case, in which boundaries must be drawn for the values of a single variable, is in fact far from simple.

Because urban reality is messy, however, combinations of characteristics often define constructs and functional areas, so that the category or interval thresholds must be defined in multivariate data space. This is not a major obstacle to simple threshold treatment as long as the set of variables is a vector and as long as the outliers in multivariate space are not theoretically interesting. For example, one may reasonably construct a weighted index of urban blight from the number or density of broken windows, graffiti, and abandoned cars; this index can be collapsed into intervals of the degree of blight. But what if the theoretically meaningful categories are complex multivariate permutations? What if we need to empirically distinguish wealthy Black suburbs from both wealthy White suburbs and moderate-income Black suburbs? Such situations require multivariate classification techniques rather than a vector of continuous variables or a single, continuous index statistic to distinguish meaningful types of variation among local areas.

The evolution of cities in late capitalism has contributed greatly to the complexity of urban processes and phenomena in ways that make multivariate category and interval permutations in variation, and hence the need for classification techniques, more important than before. Whereas enclave formation for a given group was a simple, one-directional proposition in the 1920s and 1030s, a group's enclaves in a single city may now be variously growing, disappearing, or changing hands today. Whereas the rise of suburbs in the 1950s was a straightforward process of new low-density peripheral development outside

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central city boundaries, we now must distinguish aging inner suburbs from new exurbs, and functionally suburban central city landscapes from old business districts in suburban towns. Where geographies of race were once a single-dimensional zero sum game between two groups, geographies of race and ethnicity must now accommodate combinations of four or more major identities. Classification techniques are needed to help bring order to our efforts at resolving such questions.

MULTIVARIATE CLASSIFICATION TECHNIQUES

In practice multivariate classification techniques usually mean some type of cluster analysis, although there are other approaches (about which more below). Cluster analysis refers to numerical methods for grouping objects of similar kind into categories, based on their values of multiple variables. There are many approaches to cluster analysis, the general concept—and the need for such methods—having been around for a long time. Although initially developed by a social scientist (Tryon, 1939), cluster analysis finds its most widespread scientific application today in the biological and medical sciences, including applied genomics in the service of the pharmaceutical industry, and in computer science for artificial intelligence, machine learning, and natural language processing. Other important scientific applications are in physical geography, geology, social network analysis, and data mining. Another major application of cluster analysis is in market research, notably the use of complex patterns of small-area demographic composition as proxies for consumer demand. This approach is known as geodemographics, and we shall examine in due course the debate over whether this use of cluster analysis is in fact scientific.

Cluster analysis minimizes the within-group distances (in multivariate data space) between observations and maximizes the between-group distances. There are a number of ways to specify distance in multivariate data space, the simplest being Euclidian distance between Cartesian coordinate points in j dimensions, where j is the number of variables that define the clusters. Adding to the confusion but important to the subject at hand are explicitly spatial clustering algorithms that combine geographic distance with attribute data space distance in the definition of separation between objects (Han et al., 2001). Spatial clustering techniques can, but do not always, impose contiguity constraints. Particularly where microdata are georeferenced to points, spatial clustering is one of the valuable tools available for geographic data mining and the discovery of functional territories (Miller and Han, 2001). Including geographic distance in clustering algorithms, and the additional step of enforcing contiguity, is often desirable, but not always—because the various sets of functionally similar areas in a region are themselves not always theoretically contiguous or even necessarily spatially clustered to any known degree.

There are a great many algorithms for cluster analysis, but the principal families of cluster analyses are distinguished by their starting point for grouping (Jain et al., 1999). Hierarchical (or tree-type) clustering methods start from the bottom, initially defining each observation as a cluster (distance tolerance = zero). The distance tolerance is then incrementally raised as observations are grouped at each successively smaller potential number of clusters (always written as k in cluster analysis) resulting in increasingly large clusters of observations at each decreasing level of k. Partitional clustering techniques, which include k-means clustering, are iterative techniques in which the number of clusters k is given. At the start k clusters or k initial cluster centers are randomly generated; in each iteration the algorithm re-assigns observations to clusters based on their distance from the cluster center, causing the cluster center to move. The iterations continue until no further re-assignments are made (Kaufman and Rousseeuw, 1990).

Like all empirical analysis techniques, cluster analysis has advantages and disadvantages. Its primary advantage is being the most straightforward data-driven method for creating taxonomies (classifications) of objects. The nature and meaning of the various resulting clusters are interpreted from the set of values that define the cluster centers. Often, this is a goal in itself, and thus cluster analysis is accepted as the method of choice for classification problems in many areas of the sciences. But cluster analysis is also of great value for exploratory data analysis; its use in this context can shed light on the value of various conceptual schemes for grouping objects. It can thus be an aid to hypothesis development. Moreover, the cluster membership of observations can be modeled as variables in their own right, whether as outcomes (see Reibel and Regelson, this issue) or as covariates of other outcomes (see Harris et al., 2007).

Among the disadvantages of cluster analysis is scale dependence. Although issues of scale and scope have important consequences for other methods, including linear and generalized linear modeling, the effects of changes in scale and scope on cluster analysis are more likely to cause major changes in the resulting clusters. It is therefore important not to draw broader conclusions by generalizing from cluster analyses of local areas performed on particular (e.g., metropolitan) regions or, conversely, to interpret findings from a higher scale study as either confirming or contradicting observed local patterns. Another issue is that different clustering methods will derive somewhat different clusters from the same data (Aldenderfer and Blashfield, 1984). This may seem like a serious flaw until we reflect that adding or removing an explanatory variable from a regression model or changing the link function in generalized linear models will almost always change the results of statistical modeling to some degree.

A final and potentially more serious objection to cluster analysis is that analysts often use fuzzy heuristics to determine the number of clusters to specify in their analyses. In hierarchical clustering, researchers typically look for an "elbow" where the increment in total variance combined into one fewer clusters than previously suddenly jumps. While it is possible to measure this quantity with precision, there is no basis to determine how big a jump is big enough, or which among several big jumps suggests the proper number of clusters. In recent years more robust techniques have been developed to determine the proper number of clusters that best fit the data. The general term for such methods is cluster number estimation measures; they include among other approaches cross-validation techniques (cf. Dudoit and Fridlyand, 2002; Tibshirani and Walther, 2005; Fang et al., 2007) and cluster validity indices (Maulik and Bandyopadhyay, 2002; Kryszczuk and Hurley, 2010). In cross-validation, a resampling approach is used in which data are repeatedly divided into training and test subsets and a partitional clustering method is applied to each. Comparing the degree of similarity in clustering between test and training subsets from these repeated tests yields measures of agreement that indicate the suitability of clustering schemes at each level of k. Cluster validity indices quantify properties of a clustering solution such as compactness, separation between clusters, etc. Both types of approaches provide empirically derived, comparable error measures for the various levels of k under consideration and are therefore far more robust for identifying the proper number of clusters than the heuristics of "looking for the elbow."

As mentioned above, there are other promising multivariate methods of classification besides cluster analysis that have been used in urban geography. Notably Poulsen and Johnston (2000) apply formal threshold analysis to the problem of neighborhood classification in Sydney. Threshold analysis classifies areas initially according to the binary question of whether they exceed a predetermined threshold of a single attribute. In multi-attribute threshold analysis, one takes unions of the resulting sets. This permits classification according to multivariate permutations (e.g., immigrant or non-immigrant neighborhoods with children, without children). Poulsen and Johnston establish that threshold analysis has some desirable mathematical properties lacking in cluster analysis. But the union of sets based on initial binary threshold classifications is a more rigid constraint than is imposed in cluster analysis, especially as the number of variables in the analysis rises beyond two. In addition, the technique requires theoretical justification for the selection of thresholds. In situations where theory is lacking and classification is desired as an exploratory data mining tool or aid to hypothesis development, threshold analysis is therefore less suitable than cluster analysis.

WHY HAVE CLASSIFICATION TECHNIQUES BEEN SLOW TO CATCH ON IN URBAN GEOGRAPHY?

As the preceding arguments attest, there has been a clear need for classification methods in urban geography. We might pause to ask why we are only beginning to see much application of classification methods in this context. The adoption of classification approaches in urban geography and urban studies generally seems to be hindered by three historical developments: the decline of diverse holistic approaches in favor of segregation indices; the infatuation and subsequent disenchantment with factorial ecology (which is not a classification method but is sometimes conflated with them), and the rise of cluster analysis–driven geodemographics in market research and its estrangement from academic social science.

In the course of the 20th century, urban social science in the U.S. gradually evolved from the relatively holistic descriptive-quantitative approach of the early Chicago school human ecologists (Park et al., 1925; Park, 1936) to a highly analytical approach that emphasized segregation indices (Duncan and Duncan, 1957; Taeuber and Taeuber, 1965, et seq.). That emphasis largely survives to this day, and the classical theories of human ecology have been generally discredited by a critique asserting that the theory amounts to a justification of a profoundly unfair status quo. Specifically, the human ecologists' use of the naturalistic language of biology strongly implied that the forces separating social groups spatially in cities are natural phenomena, or at least unproblematic and fair market outcomes, rather than reflecting manmade social structures, cultural norms, and institutions (Logan and Molotch, 1987; Gottdiener, 1994).[3]

[3]Another development since the late 1960s is the association of the word "ecology" in the popular imagination with the preservation or destruction of systems in the natural environment, rather than the evolutionary change dynamics of either natural or social systems. Consistent with this shift, the phrase human ecology came to refer to human impacts on natural environments rather than to the system dynamics of human communities (cf. the founding, in 1985, of the *Journal of Human Ecology*).

While the theoretical shortcomings of human ecology addressed by these authors and others are quite real, their successful critique resulted in something of a theoretical vacuum in quantitative neighborhood studies. Gradually, new theories emerged in the 1980s and 1990s to explain observed declines or persistence in levels of segregation (principally place stratification and spatial assimilation; see Charles, 2003). But the central paradox of that era of neighborhood research is the effort to explain neighborhood change over time using almost exclusively segregation statistics that cannot measure change over time directly. This has led scholars in the last decade to seek out other ways to examine neighborhood change using individual mobility, local trend data, and other alternative measures.[4]

A different kind of data reduction approach once featured prominently in urban neighborhood research. The use of factor analysis in neighborhood studies is not a classification technique and does not directly address category or interval questions regarding variation among local areas, despite the fact that conceptual similarities between the two techniques cause some to conflate them and conclude that classification has nothing to add to work done decades ago. The essential computational difference is that cluster analysis combines cases while factor analysis combines variables. The key analytic difference is that cluster analysis creates taxonomies of observations directly, while factor analysis can group observations only indirectly, incompletely, and with more undesirable constraints.

Factorial ecology goes back to Shevky and Bell (1955). During the 1960s and early 1970s, a great many articles appeared that can be grouped under the factorial ecology framework (see Berry, 1971; Hunter, 1972; Palm and Caruso, 1972). In these studies, a considerable variety of variables representing neighborhood characteristics were reduced via principal components analysis or related techniques to a manageable number of factors (thus reducing problems of multicollinearity and revealing structural relationships among neighborhood variables). The factors could be characterized in terms of the factor loadings of their individual component variables, and they were used to describe the broad categories of local and household characteristics that shaped neighborhood types. After reaching its limits as a descriptive and exploratory approach in the 1970s, factorial ecology has made a small comeback in recent years. Unlike the first-generation work, recent studies applying factorial ecology generally use factor analysis of urban neighborhoods as a means to some other theoretical or analytical end, rather than as an end in itself.[5]

Thus, among other things, factorial ecology can superficially describe *the strongest, most common* patterns of variation among neighborhoods in terms of linear covariance, which crudely implies a neighborhood taxonomy—hence the conflation with classification approaches. The problem with factor analysis as indirect classification is that infrequent (but often dramatic and theoretically important) permutations are lost in the randomness of lower order principal components. It therefore cannot be used to empirically identify the boundaries of permutation groups in multivariate data in ways that cluster analysis can. In practice, and because they operate very differently, the two techniques can be profitably combined: a large number of relevant attributes (variables) can be loaded into factors; the observations are then clustered on their scores for the derived factors.

Geodemographics has come to mean the application of small-area demographic techniques to specific applied (and typically commercial) ends—market research primarily,

[4]For a review of these developments, see Holloway et al. (2011) and Reibel and Regelson (this issue).
[5]See, for example, Morenoff and Sampson (1997), Wyly (1999), Johnston et al. (2004), and Vicino et al. (2007).

but also political campaigns (electoral and otherwise) and the siting of public or social service outlets. In the latter areas of application, geodemographics shades into practical spatial problems in urban planning and public policy, although such site selection tasks are not infrequently contracted to consulting firms whose primary business is market research. One of the largest early contracts awarded to U.S. geodemographics giant Claritas was from the National Institutes of Mental Health, to target services to the populations of Community Mental Health Centers' catchment areas (Robbin, 2009).

The primary tool of geodemographics is, not surprisingly, cluster analysis. I earlier called into question the scientific bona fides of geodemographics, despite the fact that its commercial application of cluster analysis techniques is in ways analogous to the applied science of industrial genomics. This reflects the opinions of many of my peers. In other words, many academic social scientists ignore geodemographics, which they view as overly simplistic and flawed, while tenured geneticists and bio-mathematicians are starting companies and losing their best graduate students to high-paying industrial jobs. Why the contrast?

Geodemographics has clear roots in the social science literature (cf. Bell, 1955; Kitigawa and Taueber, 1963; Howard, 1969). In the 1970s and 1980s, the early days of geodemographic marketing applications, the systems and practitioners generally had strong academic backgrounds and often retained close ties to national and local census and service planning efforts (Batey and Brown, 1995). Well into the 1980s, distinguished academic demographers were publishing research in the marketing-oriented *American Demographics* magazine (Farley and Bianchi, 1983; DeJong et al., 1987). But as the geodemographics industry achieved traction during this period, it gradually lost its connections to academic social science, mainly because of a lack of interest and understanding on the part of the academics (Openshaw, 1995).

This brings us to the major difference between urban social science and bioinformatics in terms of commercial/academic discourse. Whereas academic and industrial bioinformatics have advanced together and developed intellectual and methodological synergies, urban social science and geodemographics ultimately did not. Geodemographics quickly embraced cluster analysis; over time, without significant input from leading urban methodologists, the industry lost interest in remaining on the cutting edge of emerging robust cluster analysis techniques. Simultaneously, mainstream academics came to avoid clustering techniques in their own (scientific) work. This in turn made academics less potentially valuable to the geodemographics industry, and the feedback loop drove a wedge farther and farther between the two camps.

The different responses of these respective groups of academics have a structural component as well. Medications and gene therapies absolutely require that the best possible analytical techniques be applied in their development in order to be viable products. While geodemographics would also undoubtedly benefit in terms of power and accuracy from the application of robust clustering techniques, it is likely that in many situations the increment in quality is not sufficient to justify the cost of overcoming methodological inertia. This is a cost/benefit problem, not a scientific problem: many areas of technology and product development simply do not require and could not support the most sophisticated available scientific methods. Their use would be overkill, and this seems to explain at least in part why academic urban research and geodemographics drifted apart. It does not, however, explain why many urban social scientists appear to have turned their backs on a promising *scientific* methodology, and until recently rarely participated in advancing its frontiers, partly because related commercial applications were sometimes not as sophisticated as possible.

CATEGORIES AND CLASSIFICATION IN NEIGHBORHOOD SCALE URBAN RESEARCH

A great many studies have operationalized theoretical considerations into classifications or typologies of urban neighborhoods with the number of classes and the boundary thresholds between classes specified *a priori*. This practice dates back to Booth (1892), whose work inspired Robert Park and the Chicago ecologists. A number of these works identify distressed or impoverished neighborhoods (see, for example, Stegman, 1979; O'Laughlin, 1983; Telles, 1995). Others use empirical criteria to create functional definitions of inner city, inner suburban, and outer suburban neighborhoods (Lee and Leigh, 2005; Cooke and Marchant, 2006, Hanlon and Vicino, 2007). A third group classifies neighborhoods according to racial and ethnic composition or transition (Alba et al., 1995; Denton and Massey, 1991; Clark, 1993). However, and by definition, none of these studies using *a priori* category boundary thresholds employs classification techniques to empirically determine either the proper number of groups to analyze or the proper class boundaries for classification.

In recent years urban scholars have begun applying cluster analysis techniques to the sorts of urban questions that previously were addressed with classification schemes specified *a priori*. Geodemographics techniques have been adapted for use in scientific applied geography by Williamson et al. (2005), Harris et al. (2007), and Singleton and Longley (2009a, 2009b). Cluster analysis has also been used to good effect in the analysis of local real estate submarkets (see, for example, Goetzmann and Wachter, 1995; Hoesli et al., 1997; Jackson, 2002). There has been a group of studies analyzing suburbs using cluster techniques (Logan and Zhang, 2004; Mikelbank, 2004; Baum et al., 2006; Vicino, 2008; Hanlon, 2009).

The area of neighborhood research that is perhaps most resistant to classification techniques is neighborhood race and ethnicity (for exceptions see Modarres, 2004; Reibel and Regelson, 2007). Despite a general recognition of enclave and succession (i.e., interval) phenomena in neighborhood race and ethnic dynamics, the particular dominance of segregation indices appears to have inhibited a more rapid adoption of clustering methods in studies of neighborhood race and ethnicity. Regardless of the speed of adoption to date, investigations of all types of neighborhood-scale threshold and interval phenomena stand to benefit greatly from the application of formal classification techniques.

IN THIS ISSUE: NEW RESEARCH ON THE CLASSIFICATION OF NEIGHBORHOODS

The contributing authors in this focus section provide examples of the power of such classification techniques in neighborhood research. Brian Mikelbank performs a pooled cross-sectional cluster analysis of standardized (relative to metropolitan means) census tract–level demographic, economic, and housing data across four decades to create a combined taxonomy of relative neighborhood conditions. By examining the patterns of change over time in the conditions of individual tracts (i.e., in their cluster membership), he provides evidence regarding the life cycle of local urban areas. Of particular interest are details of the aging process of suburbs, and the implications for regional economic sustainability policies.

By combining this pseudo-survival perspective with pooled cross-sectional cluster classification, Mikelbank brings unprecedented depth and verisimilitude to the analysis of neighborhood change. Until now, as we have seen, investigators have used trend data for a single construct to measure change. Most studies of neighborhood change, whether single measure or multivariate/multigroup, have used category boundaries specified *a priori*. Even those multivariate/multigroup dynamic studies using cluster analysis have used single interval trend data, resulting in findings that are dependent on the particular time frame of the trend interval. Mikelbank's method provides a much more comprehensive, less frame dependent picture of the trajectory of local urban areas as revealed by shifts between multivariate categories that remain constant over time.

Logan et al. use point-geocoded historical microdata to advantage in their innovative approach to identifying the existence and extent of local ethnic enclaves. Their study of Newark, New Jersey in 1880 applies explicitly spatial clustering techniques to the three main ethnic groups in Newark at that time (Yankees, Germans, and Irish). Logan et al. explore various approaches to explaining the relationship between people's background characteristics and the characteristics of their residential location, and they propose criteria for establishing the boundaries of identified ethnic neighborhoods.

The use of microdata and spatial clustering enable Logan et al. to avoid the difficulties (notably the modifiable areal unit problem) arising from the use of data from pre-defined local districts. By effectively discovering enclaves through geographic pattern recognition, they provide new insight into both the limitations of pre-defined areas such as census tracts and the true geographic nature of urban enclaves and neighborhoods more generally.

Reibel and Regelson create a racial and ethnic classification of census tracts in the 50 largest U.S. metropolitan areas (MSAs/PMSAs) based on 1990 to 2000 proportional trends. Their analysis reveals highly variable patterns of neighborhood racial and ethnic change, commonly including both zones of stable integration and the expansion of segregated or segregating enclaves within the same city. This situation, which the authors term fragmented diversity, is consistent with findings in recent work by other geographers. Reibel and Regelson also find a strong multi-group stability effect; a tendency for integrating neighborhoods to include substantial numbers of multiple minorities while segregating neighborhoods tend to involve transitions predominantly between two groups. This pattern was also documented in recent work by Logan and Zhang (2010).

Reibel and Regelson also make two methodological contributions. As discussed above, scale dependency is pronounced in cluster analyses; while regional (e.g., metropolitan) case studies reveal multivariate classes at the chosen scale, they tell us nothing about higher scale patterns in the data. Studies such as this one provide meaningful results that apply to the nation as a whole. Reibel and Regelson also model the covariates of cluster membership, treating the derived clusters as multivariate trend types that are interpreted from their central data values. In this application, the initial clustering becomes a sort of data processing technique, a sophisticated recode as it were, that precedes (and makes possible) statistical analysis and hypothesis testing involving complex multivariate neighborhood types.

Vicino et al. examine the urban geography of immigrant neighborhoods in U.S. central cities. In their tract-level analysis of the immigrant population of the 18 U.S. consolidated metropolitan areas (CMSAs), they conduct a principal components analysis to determine the primary dimensions of social and economic spatial structure in central cities. Following this, they perform a cluster analysis to create a typology of immigrant neighborhoods based on characteristics of race, class, immigrant status, and housing dimensions. Like Reibel and Regelson, Vicino et al. achieve national-scale results using a different subset of metropolitan places that includes the largest city systems. Their classification of immigrant neighborhoods, while cross-sectional in time, is theory driven and comprehensive in terms of urban ecology measures. Finally, Vicino et al. demonstrate both the differences between factor analysis and clustering and the advantages of combining the two approaches.

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